

KING SHAKA INTL, South Africa

WMO: 685920

Lat: 29.602S

Lon: 31.124E

Elev: 109

StdP: 100.02

Time Zone: 2.00 (E02)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	8.8	0.9	4.1	15.9	2.8	4.7	14.7	10.4	18.1	9.6	18.3	1.2	210	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.0	30.2	22.5	29.1	22.6	28.1	22.4	24.5	28.1	24.0	27.5	23.5	26.8	5.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.2	26.5	22.9	17.9	26.2	22.2	17.1	25.6	74.8	28.2	72.5	27.7	70.5	26.8	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.4	8.4	DB	4.4	35.4	0.7	2.0	3.9	36.8	3.5	38.0	3.1	39.1	2.6	40.5	(4)
(5)			WB	3.3	26.0	0.8	0.6	2.7	26.4	2.3	26.8	1.9	27.2	1.3	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.2	23.7	23.8	23.3	20.8	19.1	16.7	16.0	17.3	18.7	19.5	21.0	22.6	(6)	
	DBStd	3.30	1.60	1.58	1.69	2.11	1.87	1.87	1.69	2.36	2.27	2.19	2.22	2.07	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	232	0	0	0	3	14	56	76	48	21	13	2	1	(9)	
	CDD10.0	3721	424	387	412	323	282	201	184	227	262	296	331	391	(10)	
	CDD18.3	911	166	153	154	76	37	6	2	17	33	51	83	133	(11)	
	CDH23.3	4616	940	893	827	356	197	91	64	127	130	157	275	559	(12)	
(13)	CDH26.7	703	137	132	115	50	27	19	11	28	31	23	56	76	(13)	
(14)	Wind	WSAvg	3.6	4.1	3.9	3.6	3.3	2.9	2.8	3.0	3.5	3.9	4.2	4.3	4.2	(14)
(15) Precipitation	PrecAvg	858	120	89	87	50	32	20	44	34	58	98	100	125	(15)	
	PrecMax	1153	301	228	205	137	144	69	193	110	200	273	263	320	(16)	
	PrecMin	422	48	15	29	9	0	0	0	3	12	21	32	41	(17)	
	PrecStd	198	57	58	52	34	35	20	61	26	39	54	51	74	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.0	30.4	30.7	30.2	29.8	29.0	28.2	30.0	29.9	29.8	31.4	30.9	(19)	
		MCWB	24.6	23.8	23.4	21.9	18.1	16.0	15.3	16.9	18.1	21.6	22.9	23.6	(20)	
	2%	DB	29.1	29.1	29.0	28.0	26.9	25.5	25.0	26.4	26.4	26.2	28.0	28.4	(21)	
		MCWB	23.7	23.7	23.3	22.1	19.5	15.7	15.7	17.5	19.7	20.6	22.5	23.3	(22)	
	5%	DB	28.1	28.1	27.9	26.2	25.2	23.9	23.1	24.2	24.2	24.9	26.0	27.1	(23)	
		MCWB	23.1	23.1	22.8	21.1	19.3	16.2	15.9	17.4	19.1	20.1	21.3	22.5	(24)	
	10%	DB	27.1	27.2	27.0	25.2	24.1	22.4	21.8	22.6	23.1	23.8	24.8	26.0	(25)	
		MCWB	22.5	22.5	22.4	20.7	19.1	16.1	16.1	17.2	18.5	19.6	20.7	21.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	25.1	24.7	23.6	22.4	19.6	18.6	20.2	21.5	22.3	23.9	25.1	(27)	
		MCDB	29.1	29.0	28.0	28.0	26.2	22.9	22.2	24.6	25.5	26.8	28.4	29.1	(28)	
	2%	WB	24.5	24.3	24.0	22.7	21.0	18.5	17.9	19.3	20.5	21.4	22.7	24.0	(29)	
		MCDB	27.9	27.9	27.4	26.4	24.6	22.4	21.7	23.7	24.4	24.9	26.6	27.4	(30)	
	5%	WB	23.7	23.7	23.5	22.0	20.2	17.7	17.3	18.5	19.7	20.7	22.0	23.1	(31)	
		MCDB	27.0	27.1	26.8	25.6	23.8	22.0	21.2	22.8	23.2	23.9	24.9	26.2	(32)	
	10%	WB	23.1	23.2	23.0	21.3	19.6	16.9	16.6	17.7	19.2	20.1	21.2	22.5	(33)	
		MCDB	26.3	26.3	26.1	24.6	23.1	21.0	20.6	21.6	22.6	23.2	24.1	25.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	7.0	7.4	8.7	10.4	12.5	11.9	10.8	8.5	7.4	6.7	6.3	(35)	
		MCDBR	8.0	8.1	8.8	10.4	12.5	15.9	15.9	14.9	11.4	9.7	9.3	7.8	(36)	
	5% WB	MCWBR	4.3	4.5	4.7	6.1	7.5	9.5	9.7	7.8	6.4	5.8	5.1	4.3	(37)	
		MCWBR	7.3	7.7	8.2	9.5	10.8	12.9	12.9	11.8	10.2	8.9	8.3	7.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.450	0.438	0.431	0.402	0.376	0.360	0.370	0.432	0.481	0.494	0.465	0.458	(40)	
		taud	2.312	2.349	2.349	2.392	2.430	2.450	2.410	2.225	2.109	2.118	2.235	2.273	(41)	
	Ebn at Noon	Ebn at Noon	896	888	859	833	807	798	804	788	798	828	876	891	(42)	
		Edh at Noon	139	130	123	106	92	86	93	124	153	162	149	145	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.94	5.83	5.08	4.12	3.53	3.14	3.35	3.91	4.50	4.91	5.41	5.87	(44)	
	RadStd	RadStd	0.47	0.43	0.26	0.25	0.18	0.15	0.22	0.23	0.29	0.43	0.46	0.43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page